

Wang Zhibin

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RESEARCH INTERESTS

UMI-style systems(mitigate the domain gap); fast world models(use latent dynamics modeling instead of explicit video generation in the world model); bridging the gap between low-level controller gains and learned high-level policies for robust action generation; Sim2Real transfer under dynamics mismatch and domain randomization; self-improving agents that leverage their own execution feedback for continual policy refinement.

EXPERIENCE

Galaxea Dynamics

Apr. 2026 – Present

VLA Pre-training and Post-training Research Intern

Shanghai, China

- Developed **robotic VLA foundation models** integrating **world-action modeling** with scalable action prediction, bridging **video-based world models** and robot action generation to improve cross-task and cross-embodiment **generalization**.
- Designed and implemented **post-training pipelines** for VLA policies, including **verification-based evaluation**, **failure-case analysis**, and **iterative data curation** to improve policy robustness with minimal human supervision.
- Investigated **test-time adaptation** strategies including **action chunk optimization**, **correlated noise scheduling**, and **delay-aware inference**, enabling policies to adapt dynamically to new tasks and reduce compounding errors.
- Built reproducible training infrastructure supporting **multi-dataset mixture**, **loss alignment verification**, and **checkpoint-level regression testing** for scalable robot learning across diverse manipulation benchmarks.
- Contributed to open-source **base model release** and **benchmark evaluation** (RoboArena-style model comparison), supporting community-driven **recursive improvement** of VLA policies through standardized evaluation protocols.

Bosch Intelligent Driving and Control Division (XC)

Oct. 2025 – Apr. 2026

Advanced Driver Assistance Dynamic Perception Algorithm Intern

Shanghai, China

- Deployed **autonomous driving perception models** on **edge hardware**, gaining deep experience in bridging neural network capabilities with real-world **hardware constraints** and **latency requirements**.
- Developed **model compression pipelines** including **graph optimization**, **operator fusion**, **mixed-precision quantization** (PTQ/QAT), achieving deployment-ready inference while preserving **perceptual accuracy**.
- Profiled and resolved **quantization-sensitive bottlenecks** through systematic **layer-wise analysis** and **precision policy search**, informing best practices for **efficient model deployment** in resource-constrained embodied systems.

Horizon Robotics

Mar. 2024 – Sep. 2024

3D Vision Algorithm Intern

Shanghai, China

- Developed **monocular 3D perception** and **camera calibration** algorithms for autonomous driving, building strong foundations in **geometric reasoning** and **2D-to-3D spatial understanding** critical for embodied agents.
- Designed **data preprocessing** and **coordinate transformation pipelines** integrating **neural predictions** with **geometric constraints** (EPnP, camera intrinsics), enabling robust **3D localization** under real-world distribution shifts.
- Conducted systematic **error diagnosis** and **corner-case mining** to identify failure modes, developing an empirical methodology for **data-driven model improvement** and **targeted data curation**.

DJI

Apr. 2023 – Aug. 2023

SLAM Algorithm Engineer Intern – Testing

Shenzhen, China

- Built a **motion capture-based ground truth pipeline** for UAV pose estimation, gaining hands-on experience in **rigid-body modeling**, **6-DoF pose acquisition**, and **trajectory evaluation** for robotic systems.
- Developed data processing tools for **time synchronization**, **coordinate frame alignment**, and **quaternion representation**, establishing rigorous evaluation methodology for **SLAM** and **localization** algorithms.
- Addressed real-world **sensor noise**, **marker occlusion**, and **pose discontinuities**, building practical intuition for **robustness challenges** in physical robot perception systems.

EDUCATION

Shanghai Maritime University

B.Eng. in Electronic Information Engineering

Sep. 2022 – Jun. 2026

Shanghai, China

TECHNICAL SKILLS

VLA Training & Post-Training: Independently built and maintained multi-GPU VLA training pipelines (DDP/FSDP); experienced in flow matching, diffusion policy training, action chunking strategies, and post-training techniques including verification-based filtering and iterative data selection.

World Model & Multimodal Reasoning: Hands-on experience integrating vision-language backbones (PaliGemma, Qwen2-VL) with action decoders; familiar with video-conditioned dynamics prediction, forward-inverse consistency objectives, and bridging visual representations with physical action spaces.

Data Pipeline & Curation: Built data pipelines over Open-X-Embodiment, LeRobot, and custom HDF5 formats at multi-dataset scale; practiced in quality filtering, cross-embodiment normalization, dataset mixing strategies, and diagnosing how data composition impacts downstream policy performance.

Simulation & Evaluation: Set up closed-loop evaluation in LIBERO, MuJoCo, and Isaac Lab; experienced in defining success metrics, analyzing failure modes, running ablations, and using simulation as a feedback loop for policy iteration and Sim2Real transfer.

Engineering: Proficient in Python/PyTorch, C/C++; comfortable with Linux, Docker, distributed debugging, model profiling (ONNX/TensorRT), and building reproducible experiment infrastructure that supports rapid research iteration.